J Mater Sci (2021) 56:3551–3561

Metals & corrosion

Oxidation resistance properties of refractory high-entropy alloys with varied AlxCrTiMo content

Ruiyang Zhang[1,2], Junhu Meng[2], Jiesheng Han[2], Kelimu Tulugan[1,*], and Rui Zhang[1,*]

[1] School of Mechanical Engineering, Xinjiang University, 830000 Urumqi, PR China
[2] Key Laboratory of Science and Technology On Wear and Protection of Materials, Lanzhou Instituteof Chemical Physics, Chinese Academy of Sciences, 730000 Lanzhou, PR China

Received: 14 July 2020
Accepted: 21 October 2020
Published online: 27 October 2020

© Springer Science+Business Media, LLC, part of Springer Nature 2020

ABSTRACT

The quaternary light refractory high-entropy alloys of $Al_xCrTiMo$ were fabricated by spark plasma sintering method. Investigation of the oxidation kinetics behavior of the alloys with different Al content at 1000 °C shows varying degrees of weight gain, and the curve of the sample with low Al concentration expresses distinctly different variation tendency. Formation process of oxide layers was analyzed and microstructure and cross sections of distribution of elements were examined by scanning electron microscope (SEM) and energy dispersive spectroscopy (EDS). TiO_2 and Al_2O_3 phases were observed on the oxide film after exposure to air at 1000 °C for 1 h and 7 h time intervals. These experiments show that the Al-content in the alloy can promote the formation of Al_2O_3, and the equilibrium of the volatilization rate of the refractory elements as well as the formation rate of the oxidation protective layer determines the weight gain of the $Al_xCrTiMo$.

Introduction

To meet the requirements of aerospace industry exploration and work in severe environment, it has been found in recent decades that high-entropy alloys (HEAs) have high-temperature properties that can be competitive with nickel-based superalloys, and have the potential to become high-temperature candidate materials [1, 2]. Different from the previous ideas of alloy design, the novel perspective proposed by Yeh et al. show that HEAs possess excellent high-temperature strength, outstanding wear resistance and corrosion resistance than traditional compound alloys, which consist of proportion of approximate compositions [3, 4]. On the other hand, the properties of different elements complement each other and accelerate the generation of exceptional performance in HEAs. The emerging multi-component alloys, therefore, have now captured much attention due to

Handling Editor: N. Ravishankar.

Address correspondence to E-mail: Kream0401@163.com; real8898963@163.com

https://doi.org/10.1007/s10853-020-05480-y

their advantages, such as better mechanical properties, corrosion resistance, wear resistance, high-temperature softening resistance and the like [5–10]. The refractory high-entropy alloy (RHEAs) is composed of some refractory elements, which continues the characteristics of refractory metals and exhibits better performance than conventional alloy composites in high-temperature environment [11]. Senkov et al. first conducted experiments on the RHEAs WNbMoTa and WNbMoTaV, which laid the foundation for the study of RHEAs [12, 13]. The results show that these two new alloys possess very high microhardness of 4.46 GPa and 5.42 GPa, yield stress at room temperature of 1058 MPa and 1246 MPa, respectively, and yield stress decreased to 405 MPa and 477 MPa at 1600 °C, respectively. Exceptional mechanical properties of these two multi-principle metallic alloys have, nevertheless, their properties should be improved so as to enhance application value attribute to their high density (12.36 and 13.75 $g{\cdot}cm^{-3}$), poor ductility at room temperature and other adverse properties. Numerous investigations have found that changes in elements or their contents of RHEAs can form a variety of alloy systems. Appropriate adjustment of alloy composition helps to obtain alloy materials with better comprehensive characteristics. In consequence, some researchers conduct design research by changing the design scheme of the alloy and adopting the method of replacing some refractory elements with appropriate elements. The reported TaNbHfZrTi, as an alloy prepared by adjusting elements, has a relatively considerable ductility ($\varepsilon > 50\%$) at room temperature [14]. The modified RHEAs not only retained single-phase body-centered cubic (BCC) crystal structure and compressive strength, but also decreased the density and hardness to 9.94 $g{\cdot}cm^{-3}$ and 3.7 GPa, respectively. The formation of this high hardness is considered to be a solid solution strengthening effect resulting from the interaction of dislocation and solute atoms [14]. In this way, light elements are introduced into the composition of refractory high-entropy alloys to achieve the purpose of reducing the density of the alloy. Commonly used low-density elements like Al and Ti. [15, 16, 21]. The addition of aluminum can support the further transformation and appearance of single-phase, and the solid solution strengthening in this process can enhance the mechanical properties [17–20]. Also, the addition of different amounts of titanium can produce large lattice distortions that can improve the strength and plasticity of alloy materials. Han et al. studied the TiNbMoTaW and TiVNbMoTaW RHEAs with the introduction of Ti, and the result shows that their compressive plastic strain at room temperature are above 10% and their room temperature yield strength increase to 1343 MPa and 1515 MPa, respectively, indicating that the introduction of Ti has an effect on the mechanical properties of the alloy [21, 22]. The addition of Cr in alloy can elevate the strength and hardness of the alloy, but introduces new phases in the alloy if added excessively, such as Laves and intermetallic compounds, which leads to a reduction in the ductility of the alloy [23]. Incorporating a lower content of non-metallic elements, such as silicon, into the multi-component alloy, maybe promotes the formation of continuous protective layer at 1000 °C and 1100 °C and also enhances the corrosion resistance to a certain extent [24–26].

While the mechanical properties are improved, the alloy added with the above-mentioned metal elements is beneficial to the formation of Al_2O_3, TiO_2, Cr_2O_3 and other oxidation protective layers, which further improves the high-temperature oxidation resistance of the alloy. The conjunction of oxidized compounds not only protects the internal substrate, but also inhibits the formation of intermetallic compound phases to some extent [27]. In the previous research, Gorr et al. adopted the blending scheme of five elements to prepare MoWAlCrTi RHEAs with preferable oxidation resistance at elevated temperature [28]. For simple phase structure, the equiatomic MoWAlCrTi, has, and the formation of dense titanium oxide and alumina oxide layers on its surface is the key to excellent oxidation resistance. Subsequently, Gorr et al. [29, 30] took Al, Cr and Ti as the substitution elements for RHEAs, and discussed the high-temperature oxidation mechanism of the new system of AlCrTiMoNb and AlCrTiMoTa. In recent years, many reasonable modification schemes and performance studies on RHEAs have been reported. Their excellent performance at high temperature has attracted much attention, but data on the high-temperature oxidation resistance of quaternary RHEAs is scarce. Therefore, the emerging RHEAs have more potential excellent performances to be explored.

In this paper, according to the design idea of element replacement, Al, Cr and Ti were used to replace the refractory elements in the previous RHEAs for composition design. The quaternary RHEAs

$Al_xCrTiMo$ (x = 0.25, 0.5, 0.75, 1) composed of three light elements and an refractory element Mo were prepared by SPS sintering. The structure and properties of alloys with different Al contents were studied, and the influence of changes in Al content on the antioxidant properties at high temperature was investigated.

Experimental procedure

The Al, Cr, Ti and Mo powder used were 99.5%, 99.5%, 99.9% and 99.95%, respectively. The mixed metal powder was prepared by planetary ball mill, which was continuously operated for 24 h at 150 $r{\cdot}min^{-1}$. The prepared mixed powder was filled into the graphite grinder and placed into the SPS furnace (SPS-20 T-10). The bulk alloy of $Al_xCrTiMo$ RHEAs with Al content of 0.25, 0.5, 0.75 and 1 were sintered in vacuum atmosphere. During the fabrication process, the furnace temperature was raised to the sintering temperature of 1400 °C with the heating rate of 40–60 $°C{\cdot}min^{-1}$, and kept warm at 30 MPa for 20 min. After sample sintering, the furnace temperature is lowered by controlling the voltage, and the sample is taken out after the temperature reduced to room temperature.

A squared block sample with the dimension of 10 mm × 10 mm × 4 mm was cut from the ingot. Then, the specimen used for the oxidation test was polished and cleaned with sandpaper and ethanol. The oxygen corrosion test was carried out in a high-temperature box-type resistance furnace in the air atmosphere at 1000 °C, and the exposure time was 1, 4 and 7 h, respectively. The heating rate is 10 $°C{\cdot}min^{-1}$. The crystal structure of $Al_xCrTiMo$ before and after high temperature oxidation was detected by Empyrean X-ray diffraction (XRD, EMPYREAN). The detection configuration was Galipixcel detector, Cu-Kα radiation (λ = 0.15405 nm) and the scanning speed was $10°{\cdot}min^{-1}$. Optical microscope (OM, Image A2m) and scanning electron microscope (SEM, JSM-5601LV) were used to investigate the microstructure and morphology of the alloy surface. The layered structure and elemental distribution of the sample cross section were observed and analyzed by energy diffraction scatter (EDS, AMETEK EDAX). In addition, the microhardness and density of the samples were measured using the Vickers hardness tester (MH-5-VM) and the Archimedes principle, respectively.

Results and discussion

Density and hardness

The density and hardness test results of $Al_xCrTiMo$ are shown in Table 1. The change of aluminum content has an effect on the density of RHEAs, which decreases with the decreasing of the Al content. The maximum and minimum alloy densities of the alloy were 6.99 $g{\cdot}cm^{-3}$ and 6.18 $g{\cdot}cm^{-3}$ with aluminum content of 1 and 0.25, respectively. The hardness of all samples are around 700 HV, which indicates that the effect of the content of Al is not obvious. Sameer Kamrudin Bachani et al. studied VNbMoTaWAl containing 2.37 at.% Al and believed that a small amount of aluminum was introduced into the RHEAs which strengthened the alloy through solid solution strengthening effect [31]. The results prove that light elements play a crucial role in lower the density of RHEAs.

Microstructure and crystal structure

The XRD comparison of the homogeneously mixed metal powder and the prepared equiatomic AlCrTiMo RHEAs sample is exhibited via Fig. 1. This phenomenon indicates that the diffraction peaks of the powder combine into a single-phase structure of the body-centered cubic (BCC), and there is no trace of intermetallic compounds in the XRD test. Different from vacuum arc melting, the alloy prepared by this method does not have metastable phase and has good stability in use [32].

The optical images of $Al_xCrTiMo$ RHEAs (x = 0.25, 0.5, 0.75, 1) are shown in Fig. 2. In the samples with different Al content, the structure of equiaxed grains are observed, but the grains sizes are different. The

Table 1 Density and hardness of the $Al_xCrTiMo$ RHEAs

Alloy	Density ($g{\cdot}cm^{-3}$)	Hardness ($HV_{0.2}$)
AlCrTiMo	6.18	692 ± 8
$Al_{0.75}CrTiMo$	6.44	702 ± 5
$Al_{0.5}CrTiMo$	6.66	719 ± 10
$Al_{0.25}CrTiMo$	6.99	730 ± 9

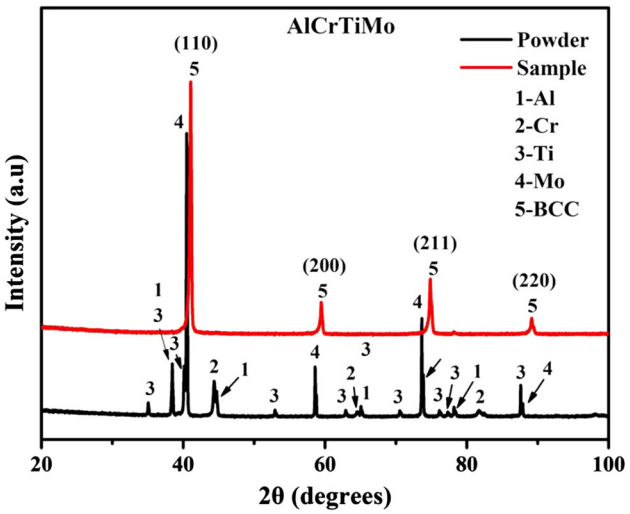


Figure 1 XRD patterns of the as-mixed powder and the as-sintered AlCrTiMo RHEAs.

grain size of the alloys with different components is shown in Table 2, and the grain sizes increases with the decreasing Al content. The structural evolution by introducing more aluminum is similar to experimental results of Liu et al. [33].

As shown in Fig. 3a, the XRD pattern expresses the crystal phase structure of $Al_xCrTiMo$ RHEAs samples fabricated with four different ingredients. Obviously, the formation of BCC single-phase structure can be observed in $Al_xCrTiMo$ (x = 0.25, 0.5, 0.75 and 1). The reduction of the aluminum content in the alloy causes the diffraction peak to shift slightly to the right, as shown in the partially enlarged view of Fig. 3b. The lattice constant of alloy is calculated by extrapolation

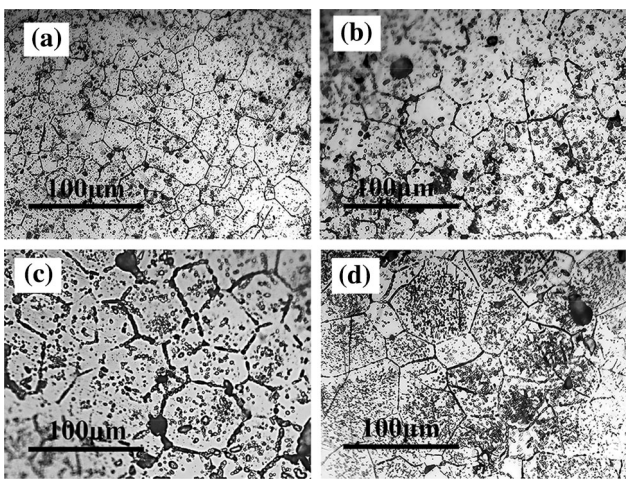


Figure 2 The optical microstructures of samples: **a** AlCrTiMo; **b** $Al_{0.75}CrTiMo$; **c** $Al_{0.5}CrTiMo$; **d** $Al_{0.25}CrTiMo$.

Table 2 Grain size and lattice constant of the $Al_xCrTiMo$

Alloy	Grain size (μm)	Lattice constant (Å)
AlCrTiMo	20	3.094
$Al_{0.75}CrTiMo$	28	3.092
$Al_{0.5}CrTiMo$	40	3.089
$Al_{0.25}CrTiMo$	48	3.088

methods, and it is affected by the change in aluminum content as shown in Table 2 [34]. The reduction in aluminum concentration slightly decreases the lattice constant. This phenomenon can be explained as that the Al atoms in the alloy have larger radius, which can produce a solid solution strengthening effect [35].

Oxidation resistance behavior

Oxidation kinetics

The oxygen corrosion behavior of Al-containing RHEAs was measured at 1000 °C air environment for 7 h. The obtained oxidation kinetic curve of Al_x-CrTiMo is shown in Fig. 4, which briefly describes the variation trend and rate of sample quality. The mass change curves of the tested alloys increased in various degrees during the initial 1 h oxidation process, and the aluminum content was positively correlated with the weight gain curve. After 1 h of high temperature oxidation, the most and least weight gains were AlCrTiMo and $Al_{0.25}CrTiMo$, respectively. The oxidation weight gain of the alloy is due to the formation and growth of the oxide layer, which is related to the concentration of the easily oxidized metal in the composition [36, 37]. In the initial 1 h oxidation process, the alloy with more aluminum content can generate an oxide protective layer at a faster rate, which provides better protection for the substrate, so the alloy has a greater weight gain. The weight gain curve of $Al_{0.25}CrTiMo$ is gentler and has the least weight gain, indicating that the rate of oxide generation is slow and the weight gain is less in this process, which can be attributed to the lower aluminum content. In addition, a verifiable explanation in RHEAs is that molybdenum easily volatilizes when the heating process exceeds 800 °C, and reacts with oxygen easily to produce MoO_3 without an oxide protective layer [38]. However, the oxide protective layer formed with less aluminum content is

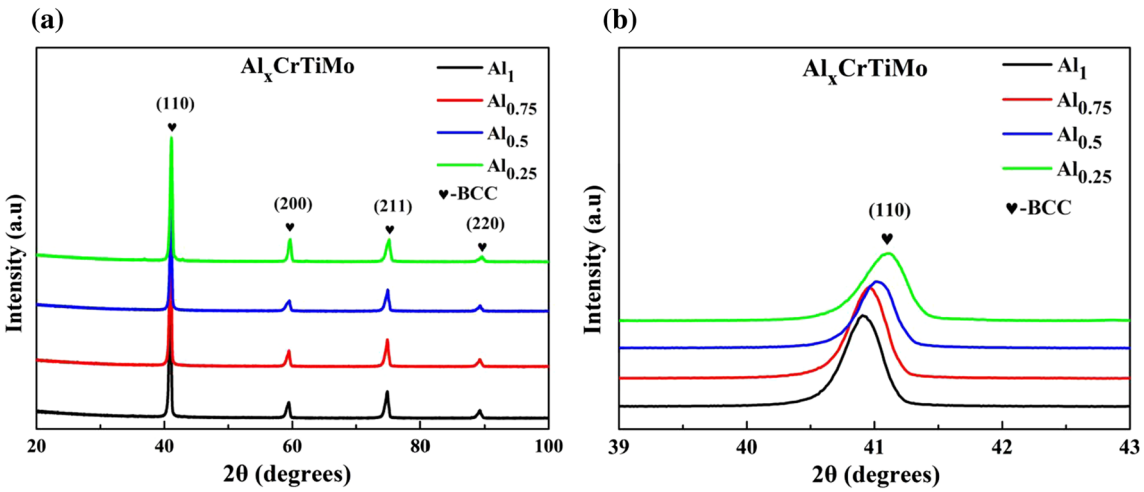


Figure 3 XRD patterns of the $Al_xCrTiMo$ RHEAs (**a**) and magnified main peaks (**b**).

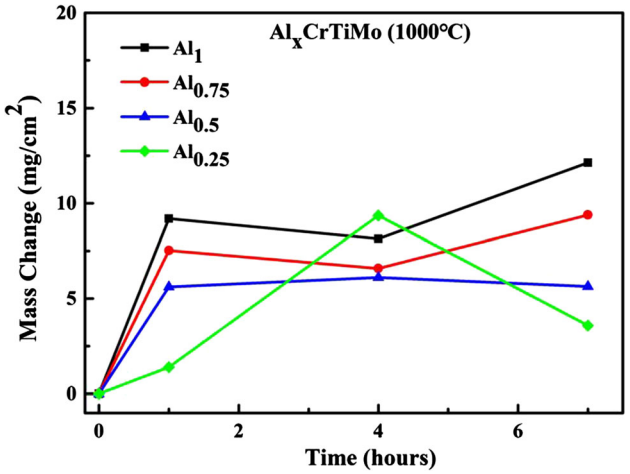


Figure 4 Mass change curves measured of the $Al_xCrTiMo$ (x = 0.25, 0.5,0.75, 1) oxidized at 1000 °C for 7 h.

not dense and discontinuous, so that the substrate of $Al_{0.25}CrTiMo$ is not well protected, and there is weight loss caused by Mo volatilization, while gaining weight. With the extension of oxidation time, protective coating is gradually formed, and the mass of specimens gain gradually.

In the experiment 1–4 h, the mass gain curves of three alloys with higher Al content showed the tendency of the slope to decline, which is due to the better protection of oxide layer. The mass gain curve of $Al_{0.25}CrTiMo$ alloy rises rapidly in this process, which may be caused by the formation of Mo-oxides gas leave porous and ruin the protective scales, increasing the oxidation of the internal metals under the oxide layer [30, 36]. In the process of oxidation for 4–7 h, the curve of the $Al_{0.25}CrTiMo$ sample fell sharply, the curve of the $Al_{0.5}CrTiMo$ sample was relatively stable, and the weight gain curves of other alloys continued to rise. A little amount of spallation and delamination was observed on the surface morphology of $Al_{0.25}CrTiMo$ alloy, and the phenomenon of negative mass growth can be explained by the continuous oxidation and volatilization of the Mo element in the substrate. In other alloys, Mo volatilization also results in weights loss, but their Al content is relatively high, and the production of more alumina is produced to increase the mass significantly, which can better guarantee the rise of weight gain curve. It is confirmed that alumina is more readily form than molybdenum oxide at 1000 °C because of the standard free energy [36].

After 7 h of high temperature oxidation, the weight gain of equiatomic AlCrTiMo RHEAs was the largest, while the weight gain of $Al_{0.25}CrTiMo$ was the smallest. During the entire oxidation process, the curves of AlCrTiMo and $Al_{0.75}CrTiMo$ keep rising. After 1 h, the curve of $Al_{0.5}CrTiMo$ tends to be flat, but the curve of $Al_{0.25}CrTiMo$ change drastically, which mainly attributed to the destruction of oxide film and the escape of excess molybdenum. From the oxidation weight gain curve, it can be concluded that the weight gain of $Al_xCrTiMo$ alloy depends on the balance between the volatilization rate of refractory elements and the formation rate of the oxidation protective layer.

Microstructural analysis of oxide scale

In order to study the high-temperature oxidation process of $Al_xCrTiMo$ HEAs, the cross sections of 1 h and 7 h samples were cut for observation and analysis. Figure 5 shows the XRD pattern of the alloy surface after 1 h oxidation at 1000 °C. TiO_2 peaks are found in all component alloys, and Diffraction peaks of Cr_2O_3 and Al_2O_3 are also found in $Al_{0.25}CrTiMo$, but only Al_2O_3 peaks are found in the other three alloys. The SEM-BEI and EDS spectra of the cross sections of the four samples after 1 h of the experiment are shown in Fig. 6, respectively. It can be observed from BSE micrograph that the thickness of oxide layer decreases with the decreasing of Al content. Therefore, the increase in oxide layer after 1 h of $Al_xCrTiMo$ with more aluminum content is the main reason for weight gain. It was observed from EDS that the main elements of the oxidation zone were Ti, Cr, Al and O. The comparison of EDS mapping of samples with different components showed that Al elements of AlCrTiMo, $Al_{0.75}CrTiMo$ and $Al_{0.5}$-CrTiMo were all clustered in the upper part of the oxidation zone, while Al distribution in $Al_{0.25}CrTiMo$ was not significantly clustered. Combined with the phase analysis of XRD and standard free energy at 1000 °C, it can be speculated that after 1 h high temperature oxidation, the outer layer of $Al_xCrTiMo$ oxide layer is TiO_2 and Al_2O_3, and the inner layer is Cr_2O_3 [36]. The oxide of Cr_2O_3 is detected in the XRD diagram and Al element aggregation in EDS mapping is not obvious, indicating that $Al_{0.25}CrTiMo$ does not form continuous and dense Al_2O_3 layer in 1 h high temperature test, leading to more molybdenum volatilization in the substrate, which is another reason for slow weight gain [36, 38]. In addition, the distribution of molybdenum and oxygen found that there was no serious oxidation in the matrix containing molybdenum.

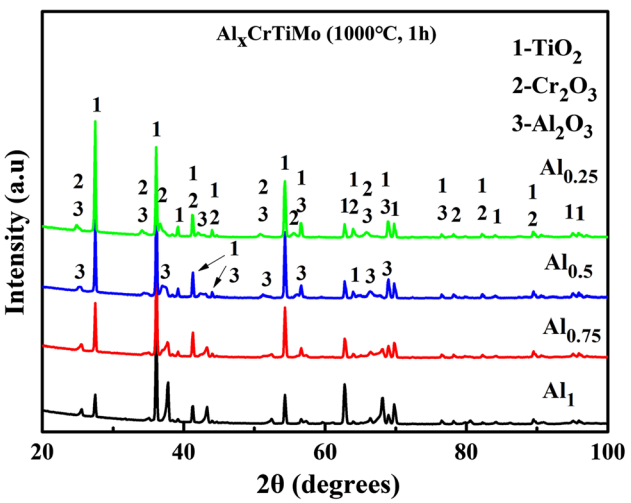


Figure 5 The XRD pattern of $Al_xCrTiMo$ (x = 0.25, 0.5, 0.75, 1) after oxidation for 1 h at 1000 °C.

The 7 h high-temperature oxidation experiment was completed, and the XRD pattern of the sample surface was shown in Fig. 7. Similar diffraction peaks of the alloys with different components were all composed of TiO_2 and Al_2O_3, indicating that a consistent oxide phase crystal structure was formed on the surface of the sample, and the oxide layer structure tended to be stable. Figure 8 shows the BSE and EDS spectra of the four alloy cross sections after the completion of the 7 h experiment. It can be seen from the BSE images that the oxide layer thickness of each sample increased significantly and the size of each alloy oxide layer did not differ significantly. The EDS atlas shows that the distribution of elements in the oxide layer of each sample is similar to the oxidation result of 1 h, in which the aggregation of Al element in the upper oxide layer increases, and Cr_2O_3 is not detected in $Al_{0.25}CrTiMo$, which may be caused by Al_2O_3 generated from above [36]. Compared with the experimental results of 1 h, the fastest and slowest growth rates of oxide layer were $Al_{0.25}CrTiMo$ and AlCrTiMo, respectively. The Al_2O_3 layer is formed faster in the alloy with higher Al content, which provides protection for the substrate earlier and makes the alloy continue to gain weight during oxidation. However, due to the insufficient aluminum content, the Al_2O_3 protective layer of $Al_{0.25}CrTiMo$ is formed slowly and not dense and path the way for the matrix reacts with oxygen, which is the main reason for its low oxidation weight gain. Comparing the oxidation results of 1 h and 4 h and different aluminum contents, it is not difficult to conclude that aluminum oxide protective layer is crucial to the oxidation resistance of $Al_xCrTiMo$ RHEAs.

After the completion of 7 h oxidation test, the thickness of oxide layer increase with varying degrees in $Al_xCrTiMo$. The alloy with 1–0.25 ratio mole Al content has average thickness of ~ 755 μm, ~ 875 μm, ~ 921 μm and ~ 1018 μm, respectively. The element distribution of the oxide coating and partial substrate is detected by EDS line scan (Fig. 9). The oxide layers of alloys with different aluminum content have similar element content

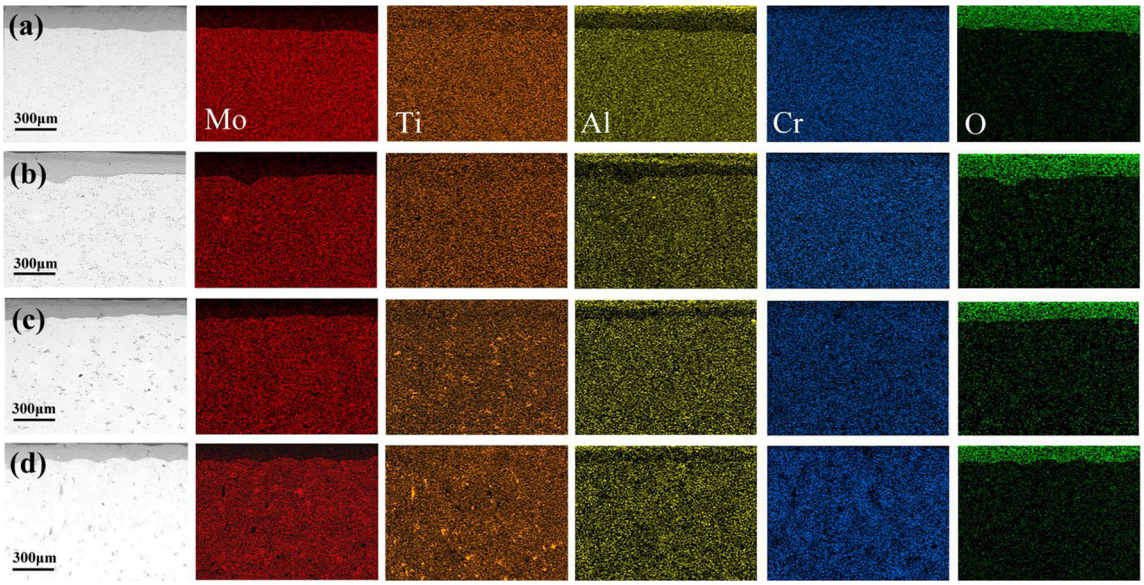


Figure 6 SEM-BEI and the corresponding EDS elemental mappings of the $Al_xCrTiMo$ cross section after oxidation for 1 h at 1000 °C. **a** AlCrTiMo; **b** $Al_{0.75}CrTiMo$; **c** $Al_{0.5}CrTiMo$; **d** $Al_{0.25}CrTiMo$.

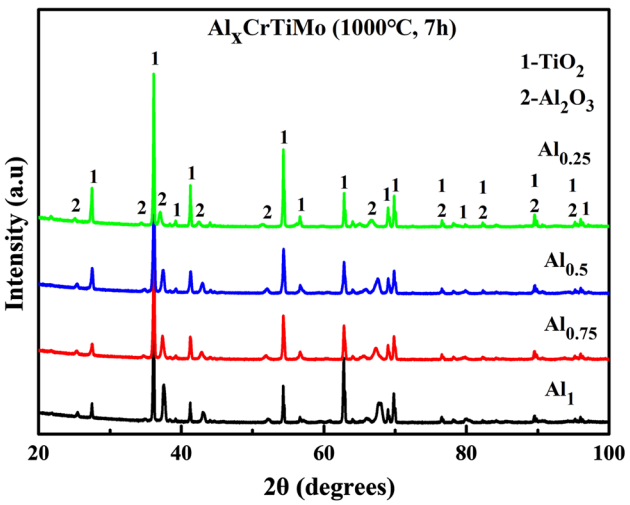


Figure 7 The XRD pattern of $Al_xCrTiMo$ (x = 0.25, 0.5, 0.75, 1) after oxidation for 7 h at 1000 °C.

distribution, the stable concentration of Al, Ti and Cr indicates that change of aluminum content in Al_x-CrTiMo has little influence on the oxide structure. The diffusion of oxygen atoms in the oxide layer resulted in the significant decrease in Mo content, which is attributed to the volatilization of Mo at high temperature [36, 38, 39]. Different with the traditional alloy in oxidation test, the weight loss of RHEAs introduced with Mo affects the weight gain curve, but the oxide layer thickness and EDS clearly indicate the antioxidant performance. The rapid formation of AlCrTiMo oxide layer and less Mo volatilization result in the largest mass increase, but its oxide layer is the thinnest among $Al_xCrTiMo$. Therefore, it can be used as an alternative material for light short-term high-temperature oxidation resistance refractory alloy.

4. Conclusion

In this study, the quaternary $Al_xCrTiMo$ RHEAs consist of different ingredients were prepared successfully by SPS technology. The effects of aluminum content on oxidation behavior of $Al_xCrTiMo$ high entropy alloys (x = 0.25, 0.5, 0.75, 1) were investigated. The results consider that low density and good oxidation resistance in short time at elevated temperature excepted to combine. The specific summaries can be stated as follows. This study provides a reference for the study of high temperature performance of light refractory high-entropy alloy.

1. Although quaternary RHEAs with different content of Al element, single BCC structure exists as equiaxed crystal in microstructure by XRD analysis. The content of Al increased from 0.25 to 1, and the alloy density decreased from 6.99 to 6.18 g·cm^{-3}.
2. The oxidation kinetics exhibits moderate mass gain in Al-cotaining RHEAs with content 0.5 to 1

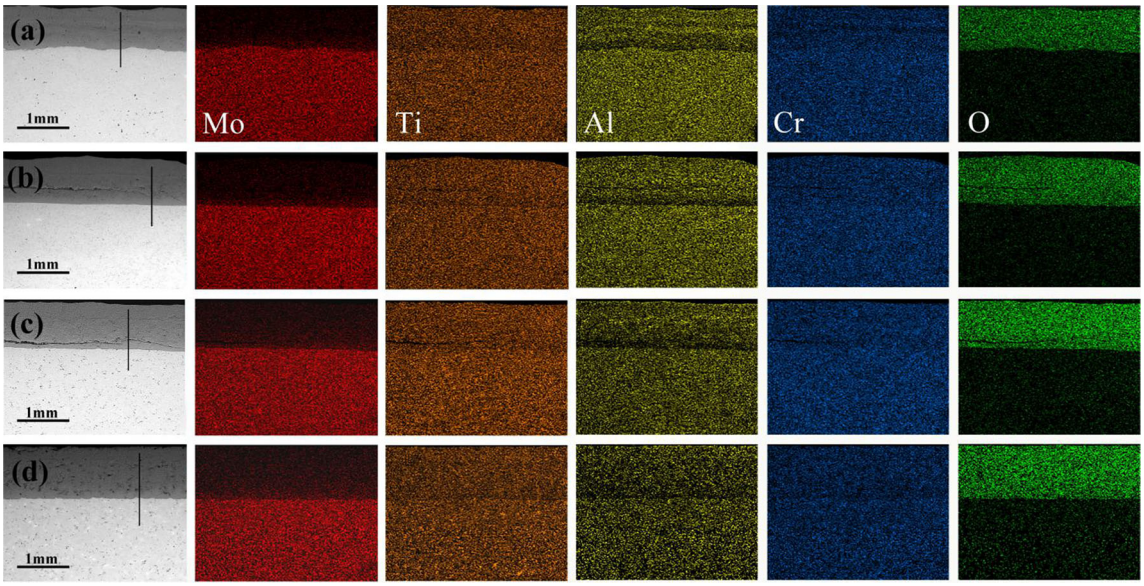


Figure 8 SEM-BEI and the corresponding EDS elemental mappings of the $Al_xCrTiMo$ cross section after oxidation for 7 h at 1000 °C. **a** AlCrTiMo; **b** $Al_{0.75}CrTiMo$; **c** $Al_{0.5}CrTiMo$; **d** $Al_{0.25}CrTiMo$.

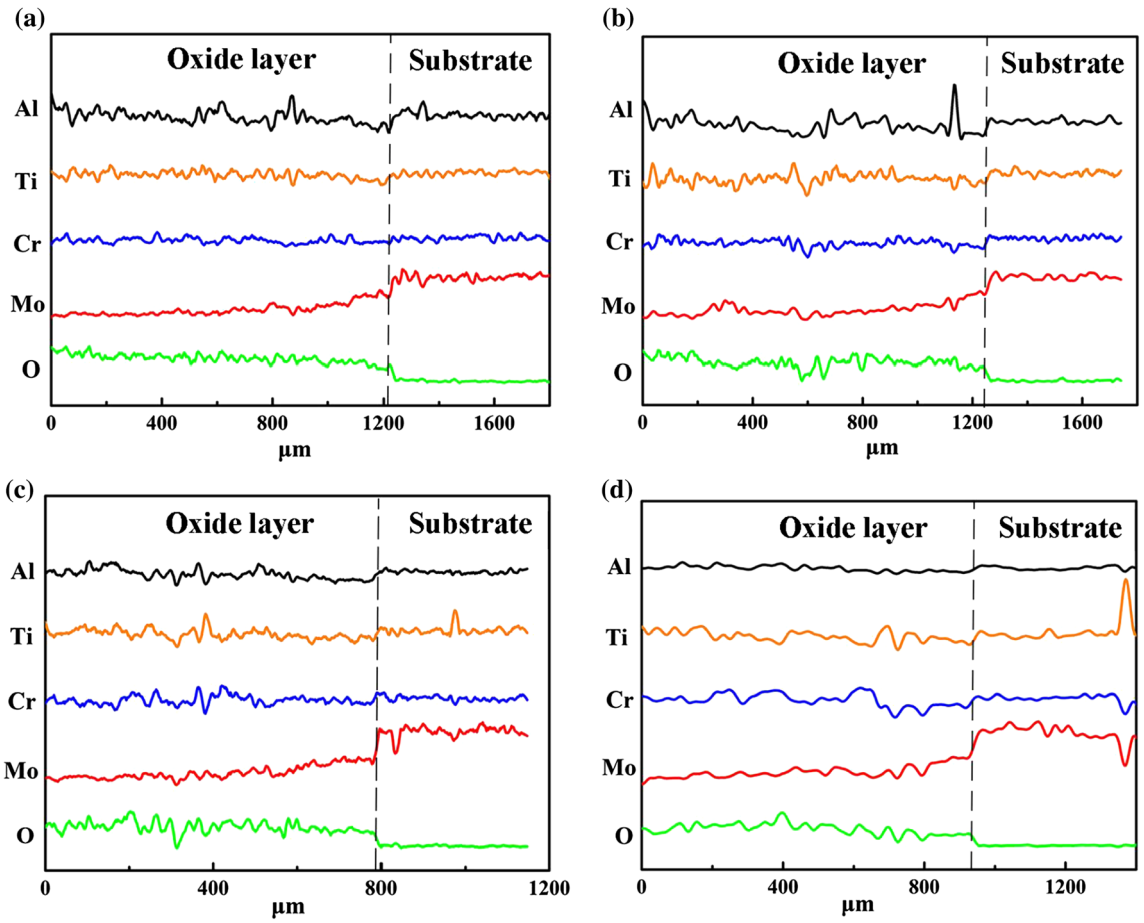


Figure 9 Chemical analyses by EDS line scan along the black line shown in Fig. 8. **a** AlCrTiMo; **b** $Al_{0.75}CrTiMo$; **c** $Al_{0.5}CrTiMo$; **d** $Al_{0.25}CrTiMo$.

during the process of oxygen corrosion, while the reduction of mass increase rate occurs twice during the oxidation process of $Al_{0.25}CrTiMo$. This first change in behavior can be possibly explained by lack of protective alumina oxide layers formation, and another change could attributed on few spallation and delamination occur on the surface of low Al-containing specimens. The oxidation layers consist of TiO_2, Al_2O_3 and Cr_2O_3 from exterior to interior of alloy orderly. High Al content alloys generate protective easily in short period, therefore, leading to good oxidation resistance at elevated temperature.

3. It is significant that Al content play a crucial role in oxidation resistant property of RHEAs, which has been confirmed again. Meanwhile, the novel composites shown promising potential in elevated temperature, and decent oxygen corrosion property of $Al_xCrTiMo$ path the way for subsequent research.

Acknowledgements

This work was supported by a grant from Tianchi Doctoral Program in Xinjiang Uygur Autonomous Region.

Author contributions

RZ: Conceptualization, Formal analysis, Data curation, Investigation, Writing—original draft, Writing—review & editing. JM: Funding acquisition, Investigation, Methodology. JH: Software, Supervision, Validation. KT: Writing—review & editing. RZ: Formal analysis, Methodology, Project administration.

Compliance with ethical standard

Conflict of interest All authors of this manuscript have directly participated in planning, execution and analysis of this study. The contents of this manuscript have not been copyrighted or published previously. The contents of this manuscript are not now under consideration for publication elsewhere. The contents of this manuscript will not be copyrighted, submitted or published elsewhere, while acceptance by Journal of Materials Science is under consideration. There are no directly related manuscripts or abstracts, published or unpublished, by any authors of this manuscript. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

References

[1] Zhang Y, Zuo TT, Tang Z, Gao MC, Dahmen KA, Liaw PK, Zhao Ping Lu (2014) Microstructures and properties of high-entropy alloys. Prog Mat Sci 61:1–93. https://doi.org/10.1016/j.pmatsci.2013.10.001

[2] Senkov ON, Woodward C, Miracle DB (2014) Microstructure and properties of aluminum-containing refractory high-entropy alloys. JOM 66:2030–2042. https://doi.org/10.1007/s11837-014-1066-0

[3] Yeh J-W, Chen S-K, Lin S-J, Gan J-Y, Chin T-S, Shun T-T, Tsau C-H, Chang S-Y (2004) Nanostructured high-entropy alloys with multi-principal elements: novel alloy design concepts and outcomes. Adv Eng Mat 6(299):303. https://doi.org/10.1002/adem.200300567

[4] Tsai M-H, Yeh J-W (2014) High-entropy alloys: a critical review. Mat Res Lett 2:107–123. https://doi.org/10.1080/21663831.2014.912690

[5] Jensen JK, Welk BA, Williams REA, Sosa JM, Huber DE, Senkov ON, Viswanathan GB, Fraser HL (2016) Characterization of the microstructure of the compositionally complex alloy $Al_1Mo_{0.5}Nb_1Ta_{0.5}Ti_1Zr_1$. Scr Mat 121:1–4. https://doi.org/10.1016/j.scriptamat.2016.04.017

[6] Senkov ON, Isheim D, Seidman DN, Pilchak AL (2016) Development of a refractory high entropy superalloy. Entropy 18:102. https://doi.org/10.3390/e18030102

[7] Jayaraja J, Thinaharan C, Ningshen S, Mallika C, Kamachi Mudali U (2017) Corrosion behavior and surface film characterization of TaNbHfZrTi high entropy alloy in aggressive nitric acid medium. Intermetallics 89:123–132. https://doi.org/10.1016/j.intermet.2017.06.002

[8] Li J, Yang X, Zhu R, Zhang Y (2014) Corrosion and serration behaviors of $TiZr_{0.5}NbCr_{0.5}V_xMo_y$ high entropy alloys in aqueous environments. Metals 4:597–608. https://doi.org/10.3390/met4040597

[9] Poulia A, Georgatis E, Lekatou A, Karantzalis AE (2016) Microstructure and wear behavior of a refractory high entropy alloy. Int J Refract Met Hard Mat 57:50–63. https://doi.org/10.1016/j.ijrmhm.2016.02.006

[10] Mathiou C, Poulia A, Georgatis E, Karantzalis AE (2018) Microstructural features and dry-sliding wear response of MoTaNbZrTi high entropy alloy. Mat Chem Phys

210:126–135. https://doi.org/10.1016/j.matchemphys.2017.08.036

[11] Senkov ON, Miracle DB, Chaput KJ, Couzinie J-P (2018) Development and exploration of refractory high entropy alloys—A review. J Mat Res 19:3092–3128. https://doi.org/10.1557/jmr.2018.153

[12] Senkov ON, Wilks GB, Miracle DB, Chuang CP, Liaw PK (2010) Refractory high-entropy alloys. Intermetallics 18:1758–1765. https://doi.org/10.1016/j.intermet.2010.05.014

[13] Senkov ON, Wilks GB, Scott JM, Miracle DB (2011) Mechanical properties of $Nb_{25}Mo_{25}Ta_{25}W_{25}$ and $V_{20}Nb_{20}Mo_{20}Ta_{20}W_{20}$ refractory high entropy alloys. Intermetallics 19:698–706. https://doi.org/10.1016/j.intermet.2011.01.004

[14] Senkov ON, Scott JM, Senkova SV, Miracle DB, Woodward CF (2011) Microstructure and room temperature properties of a high-entropy TaNbHfZrTi alloy. J Alloys Compd 50:96043–96048. https://doi.org/10.1016/j.jallcom.2011.02.171

[15] Wang W-R, Wang W-L, Wang S-C, Tsai Y-C, Lai C-H, Yeh J-W (2012) Effects of Al addition on the microstructure and mechanical property of $Al_xCoCrFeNi$ high-entropy alloys. Intermetallics 26:44–51. https://doi.org/10.1016/j.intermet.2012.03.005

[16] Senkov ON, Senkova SV, Woodward C (2014) Effect of aluminum on the microstructure and properties of two refractory high-entropy alloys. Acta Mater 68:214–228. https://doi.org/10.1016/j.actamat.2014.01.029

[17] Yang X, Zhang Y, Liaw PK (2012) Microstructure and compressive properties of $NbTiVTaAl_x$ high entropy alloys. Procedia Eng 36:292–298. https://doi.org/10.1016/j.proeng.2012.03.043

[18] Lin C-M, Juan C-C, Chang C-H, Tsai C-W, Yeh J-W (2015) Effect of Al addition on mechanical properties and microstructure of refractory $Al_xHfNbTaTiZr$ alloys. J Alloys Compd 624:100–107. https://doi.org/10.1016/j.jallcom.2014.11.064

[19] Liu CM, Wang HM, Zhang SQ, Tang HB, Zhang AL (2014) Microstructure and oxidation behavior of new refractory high entropy alloys. J Alloys Compd 583:162–169. https://doi.org/10.1016/j.jallcom.2013.08.102

[20] Chen SY, Yang X, Dahmen KA, Liaw PK, Zhang Y (2014) Microstructures and crackling noise of $Al_xNbTiMoV$ high entropy alloys. Entropy 16:870–884. https://doi.org/10.3390/e16020870

[21] Han ZD, Luan HW, Liu X, Chen N, Li XY, Shao Y, Yao KF (2018) Microstructures and mechanical properties of TixNbMoTaW refractory highentropy alloys. Mat Sci Eng A 712:380–385. https://doi.org/10.1016/j.msea.2017.12.004

[22] Han ZD, Chen N, Zhao SF, Fan LW, Yang GN, Shao Y, Yao KF (2017) Effect of Ti additions on mechanical properties of NbMoTaW and VNbMoTaW refractory high entropy alloys. Intermetallics 84:153–157. https://doi.org/10.1016/j.intermet.2017.01.007

[23] Stepanov ND, Yu Yurchenko N, Skibin DV, Tikhonovsky MA, Salishchev GA (2015) Structure and mechanical properties of the $AlCr_xNbTiV$ (x=0, 0.5, 1, 1.5) high entropy alloys. J Alloys Compd 652:266–280

[24] Gorr B, Mueller F, Christ H-J, Mueller T, Chen H, Kauffmann A, Heilmaier M (2016) High temperature oxidation behavior of an equimolar refractory metal-based alloy 20Nb-20Mo-20Cr-20Ti-20Al with and without Si addition. J Alloys Compd 688:468–477. https://doi.org/10.1016/j.jallcom.2016.07.219

[25] Müller F, Gorr B, Christ H-J, Chen H, Kauffmann A, Heilmaier M (2018) Effect of microalloying with silicon on high temperature oxidation resistance of novel refractory high-entropy alloy Ta-Mo-Cr-Ti-Al. Mat High Temp 35:168–176. https://doi.org/10.1080/09603409.2017.1389115

[26] Guo Z, Zhang A, Han J, Meng J (2019) Effect of Si additions on microstructure and mechanical properties of refractory NbTaWMo high-entropy alloys. J Mater Sci 54:5844–5851. https://doi.org/10.1007/s10853-018-03280-z

[27] Young DJ (2016) High temperature oxidation and corrosion of metals, 2nd edn. Elsevier Science, Amsterdam

[28] Gorr B, Azim M, Christ H-J, Mueller T, Schliephake D, Heilmaier M (2015) Phase equilibria, microstructure, and high temperature oxidation resistance of novel refractory high-entropy alloys. J Alloys Compd 624:270–278. https://doi.org/10.1016/j.jallcom.2014.11.012

[29] Gorr B, Mueller F, Christ HJ, Chen H, Kauffmann A, Schweiger R, Szabó DVH, M, (2018) Development of oxidation resistant refractory high entropy alloys for high temperature applications: recent results and development strategy. TMS annual meeting & exhibition. Springer, Cham, pp 647–659

[30] Gorr B, Müller F, Azim M, Christ H-J, Müller T, Chen H, Kauffmann A, Heilmaier M (2017) High-temperature oxidation behavior of refractory high-entropy alloys: effect of alloy composition. Oxid Met 88:339–349. https://doi.org/10.1007/s11085-016-9696-y

[31] Bachani SK, Wang C-J, Lou B-S, Chang L-C, Lee J-W (2020) Microstructural characterization, mechanical property and corrosion behavior of VNbMoTaWAl refractory high entropy alloy coatings: effect of Al content. Surf Coat Technol 403:126351. https://doi.org/10.1016/j.surfcoat.2020.126351

[32] Torralba JM, Alvaredo P, García-Junceda A (2019) High-entropy alloys fabricated via powder metallurgy. Powder

Metall, A critical review. https://doi.org/10.1080/00325899.2019.1584454

[33] Liu YY, Chen Z, Chen YZ, Shi JC, Wang ZY, Wang S, Liu F (2019) Effect of Al content on high temperature oxidation resistance of Al_xCoCrCuFeNi high entropy alloys (0,0.5, 1.5, 2). Vacuum. https://doi.org/10.1016/j.vacuum.2019.108837

[34] Nelson JB, Riley DP (1945) An experimental investigation of extrapolation methods in the derivation of accurate unit-cell dimensions of crystals. Proc Phys Soc Lond 57:160–177. https://doi.org/10.1088/0959-5309/57/3/302

[35] Chen H, Kauffmann A, Laube S, Choi I-C, Schwaiger R, Huang Y, Lichtenberg K et al (2018) Contribution of lattice distortion to solid solution strengthening in a series of refractory high entropy alloys. Metall Mat Trans A 49:772–781. https://doi.org/10.1007/s11661-017-4386-1

[36] Müller F, Gorr B, Christ H-J, Müller J, Butz B, Chen H, Kauffmann A, Heilmaier M (2019) On the oxidation mechanism of refractory high entropy alloys. Corros Sci 159:108161. https://doi.org/10.1016/j.corsci.2019.108161

[37] Tod, d M. Butler, Mark L. Weaver, (2016) Oxidation behavior of arc melted AlCoCrFeNi multi-component high-entropy alloys. J Alloys Compd 674:229–244. https://doi.org/10.1016/j.jallcom.2016.02.257

[38] Gulbransen EA, Andrew KF, Brassart FA (1963) Oxidation of molybdenum 550 °C to 1700 °C. J Electrochem Soc 110:952–959. https://doi.org/10.1016/0042-207X(64)91551-9

[39] Li L-C, Li M-X, Liu M, Sun B-Y, Wang C, Huo J-T, Wang W-H, Liu Y-H (2020) Enhanced oxidation resistance of MoTaTiCrAl high entropy alloys by removal of Al. SCI CHINA Mat. https://doi.org/10.1007/s40843-020-1332-2

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.